TRACK: UX/UI DESIGN · WEEK 2 · FOUNDATIONS

UX-02: Research Sprint with AI Synthesis

10 Primary Sources · Claude 3.5 Synthesis · Verification Audit · One-Page Strategic Summary

Candidate: K. M. Siam
Project: FlyCommerce OS

TOTAL DESK SOURCES	USER REVIEWS ANALYZED	SYNTHESIS ENGINE	AUDIT RESULT
10 Documented Sources	22 Real Operator Reviews	Claude 3.5 Sonnet (XML)	3 Hallucinations Corrected

1. Executive One-Page Research Summary: Top 5 Source-Tied Findings

1. Silent Sync Latency (>90m) Directly Drives Marketplace Algorithmic Demotion

McKinsey (Source 04) & /r/shopify (Source 06)

Evidence: McKinsey revealed 73% of mid-market DTC brands experienced search rank demotions or account penalties from sync latency. On Reddit, operators reported TikTok Shop webhooks take up to 3 hours to sync during sales, resulting in 200+ overbooked orders.

> **Actionable Design Mandate:** Persistent *Real-Time Sync Health Indicator* in the global nav. If webhook latency exceeds 60s, trigger an amber warning with automated inventory buffer locking.

2. "Explainable AI (XAI)" Is Required to Break the Fragile Excel Dependency

Gartner OMS (Source 05) & /r/ecommerce (Source 10)

Evidence: Gartner found 68% of advanced inventory tools are abandoned within 90 days due to "black-box complexity." Over 60% of operators keep parallel Excel sheets because they refuse to trust unexplained AI reorder numbers.

> **Actionable Design Mandate:** Contextual *Inspectable Math Drawer* on every suggested reorder showing: `(Daily Velocity × Lead Time) + Safety Buffer = PO`, paired with a 1-click CSV export button.

3. Data Formatting Clashes Paralyze Warehouse Floor Execution

ShipBob Teardown (Source 02) & /r/FulfillmentByAmazon (Source 07)

Evidence: 3PL floor operations halt when client CSV uploads have trailing spaces or mismatched dimensions, resulting in carrier cutoff misses and Amazon Inbound Placement fee penalties.

> **Actionable Design Mandate:** Client-side *Pre-Flight CSV Linting* on upload modals with 1-click "Auto-Clean & Format" and high-contrast status tokens (`STAGED`, `PICKING`, `BLOCKED`).

4. Factory Blackouts Are the Blind Spot of Reorder Automation

Shopify App Store (Source 09) & Katana Teardown (Source 03)

Evidence: Automated tools assume static lead times and fail to anticipate predictable factory shutdowns (e.g. Lunar New Year, factory retooling, canal freight disruptions).

> **Actionable Design Mandate:** Visual *Supplier Blackout Calendar* in vendor settings that automatically inflates safety stock calculations 30 days prior to supplier downtime.

5. High-Density UI Must Prioritize Status Signifiers Over Deep Navigation

Shopify Plus (Source 01) & ShipBob (Source 02)

Evidence: Shopify requires 7 clicks to transfer inventory buffers between locations; existing WMS portals bury critical stockout statuses in low-contrast monochromatic grey tables.

> **Actionable Design Mandate:** Unified high-density grid (1440px desktop / 1024px tablet) utilizing 4 high-contrast semantic tokens: `SYNCED (Green)`, `AT RISK (Amber)`, `BLOCKED (Red)`, `TRANSFER (Blue)`.

2. AI Synthesis Prompt & Human Verification Audit

```
<context>Staff Product Researcher evaluating desk research for FlyCommerce OS.</context>
<instructions>Analyze corpus of 10 sources (3 teardowns, 2 reports, 22 user reviews). Extract top themes, contradictions, and unmet needs with direct attribution.</instructions>
```

Area / Claim	Claude 3.5 Sonnet Raw Output	Human Verification & Correction
McKinsey Loss Metrics	Claude claimed: *"95% of brands lose $1M+ annually due to inventory sync delays."*	**Corrected:** McKinsey reported **73% of brands** experienced listing demotions or account penalties. The $1M+ figure was fabricated by Claude and eliminated.
Warehouse Hardware	Claude suggested: *"Operators demand iOS LiDAR mobile apps to count inventory."*	**Corrected:** Verified against ShipBob teardown and interviews: 3PLs use **Zebra/Honeywell Android scanners** and cart-mounted iPads needing 48px touch targets.
Interaction Model	Claude proposed: *"Build a conversational AI chatbot that explains recommendations."*	**Corrected:** Replaced conversational chatbot (adds unnecessary friction) with an **inline Inspectable Math Tooltip** showing exact reorder formulas on hover/click.

3. Primary Source Attribution Index (10 Sources)

Source	Type	Key Finding / Evidence
01. Shopify Inventory Matrix	Competitor Teardown	High polish but requires 7 clicks to transfer buffers; lacks real-time velocity correlation.
02. ShipBob WMS Portal	Competitor Teardown	Opaque CSV error messaging during bulk uploads; monochromatic status tags fail visual hierarchy.
03. Katana Cloud Manufacturing	Analog Teardown	Complex multi-level Bill of Materials (BOM) overwhelms solo DTC operators with enterprise jargon.
04. McKinsey Logistics Report	Industry Benchmark	73% of mid-market DTC brands suffered listing demotions from sync latency exceeding 90 minutes.
05. Gartner OMS Guide (2025)	Industry Benchmark	68% of advanced inventory tools abandoned within 90 days due to "black-box" algorithmic complexity.
06. Reddit /r/shopify	User Community Review	TikTok Shop webhook latency (up to 3 hrs) caused 214 overbooked orders and health score crash.
07. Reddit /r/FulfillmentByAmazon	User Community Review	Amazon Inbound Placement fee penalties incurred when box dimensions mismatch manifests by 0.5".
08. G2 Inventory Sync Reviews	Verified SaaS Reviews	Silent webhook failures during promotional flash sales without dashboard alert notifications.
09. Shopify App Store (Stocky)	User App Review	Static reorder formulas fail to account for supplier holiday blackouts (e.g. Lunar New Year).
10. Reddit /r/ecommerce	Operator Survey	60%+ of founders maintain duplicate Excel sheets to guard against silent SKU variant corruption.

Master Figma Deliverable (View Access): https://www.figma.com/file/r7K8x9pL4mN2qS/FlyRank-UX-UI-Design-Capstone?type=design&node-id=0%3A1&mode=design&t=review-access
GitHub Repository: https://github.com/KMSiam/Flyrank-Internship-UX-UI-Design